



CHAPTER

5

Thermodynamics

Sr. No.	Concept	Formulas	Other information
1.	Thermodynamic processes: (a) Isothermal process (b) Isochoric process (c) Isobaric process (d) Adiabatic process	$T = \text{constant}, dT = 0, \Delta T = 0$ $V = \text{constant}, dV = 0, \Delta V = 0$ $P = \text{constant}, dP = 0, \Delta P = 0$ $q = 0$	$T = \text{Temperature}$ $V = \text{Volume}$ $P = \text{Pressure}$ $q = \text{Heat}$
2.	Internal energy(U)	For ideal gas: $\Delta U = nC_V\Delta T$ For chemical reactions: $\Delta U = U_{\text{products}} - U_{\text{reactants}}$	$\Delta U = \text{Change in internal energy}$ $n = \text{No. of moles}$ $C_V = \text{Heat capacity at constant volume}$ $\Delta T = \text{Change in temperature}$
3.	Heat(q)	$q = C\Delta T$ $q = C_m\Delta T$	$C = \text{Heat capacity}$ $C_m = \text{Molar heat capacity}$
4.	Work (W): (a) Reversible process (b) Irreversible process	$W = - \int_{V_1}^{V_2} PdV$ $W = -P_{\text{ext}} \times \Delta V$	$P = \text{Pressure}$ $V_1 = \text{Initial volume}$ $V_2 = \text{Final volume}$ $W = \text{Work done}$ $\Delta V = \text{Change in volume}$ $P_{\text{ext}} = \text{External pressure}$
5.	Ist Law of Thermodynamics	$\Delta U = q + W$	$\Delta U = \text{Change in internal energy}$ $q = \text{Heat}$ $W = \text{Work}$

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6.	Work done in various process:		
	(a) Isochoric process	$W = 0, \Delta U = q = nC_V\Delta T$	W = Work done ΔU = Change in internal energy q = Heat n = No. of moles C_V = Heat capacity at constant volume ΔT = Change in temperature
	(b) Isobaric process	$W = -P_{\text{ext}}(V_2 - V_1)$	P_{ext} = External pressure V_1 = Initial volume V_2 = Final volume
	(c) Isothermal process	$\Delta T = 0, \Delta U = 0; q = -W$	
	(d) Adiabatic process	$q = 0, \Delta U = W = nC_V\Delta T$ $W = nC_V(T_2 - T_1)$ $W = \frac{nR}{(\gamma - 1)}(T_2 - T_1)$	$\gamma = \frac{C_P}{C_V}$ T_1 = Initial temperature T_2 = Final temperature
	(e) Reversible isothermal process	$W_{\text{rev,iso}} = -nRT\ln(V_2/V_1) = -nRT\ln(P_1/P_2)$	R = Gas constant P_1 = Initial pressure P_2 = Final pressure
	(f) Irreversible isothermal process	$W = -P_{\text{ext}}[nRT/P_2 - nRT/P_1]$	
	(g) Reversible adiabatic process	$PV^\gamma = \text{constant (ideal gas)}$ $TV^{\gamma-1} = \text{constant (ideal gas)}$	
7.	Enthalpy	$H = U + PV$ $\Delta H = \Delta U + \Delta(PV)$ $\Delta H = \Delta U + (P_2V_2 - P_1V_1)$ $\Delta H = \Delta U + P\Delta V + V\Delta P$ At constant pressure $\Delta H = \Delta U + P\Delta V$ $\Delta H = \Delta U + \Delta n_g RT$ (for chemical reaction)	H = Enthalpy U = Internal energy ΔU = Change in internal energy P_1 = Initial pressure P_2 = Final pressure V_1 = Initial volume V_2 = Final volume ΔV = Change in volume ΔP = Change in pressure $\Delta n_g = n_2 - n_1$ = No. of moles of gaseous products - No. of moles of gaseous reactants

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8.	Relation between C_p & C_v for an ideal gas equation	$C_p = \Delta H / \Delta T$ $C_v = \Delta U / \Delta T$ $C_p - C_v = R$ $C_p / C_v = \gamma$	C_v = Heat capacity at constant volume C_p = Heat capacity at constant pressure R = Gas constant γ = Poisson ratio
9.	Thermochemical reactions: (a) Endothermic reactions (b) Exothermic reactions	$\Delta H = H_p - H_R = +Ve$ $\Delta H = H_p - H_R = -Ve$	H_p = Enthalpy of products H_R = Enthalpy of reactants
10.	Heat of reactions / Enthalpy of reactions: (a) Heat of formation (b) Heat of combustion (c) Heat of neutralisation (d) Bond enthalpy (e) Heat of solution (f) Hess law of constant heat summation	$\Delta H = (\sum \Delta H_f)_{\text{Product}} - (\sum \Delta H_f)_{\text{reactant}}$ $\Delta H = (\sum \Delta H_{\text{comb.}})_{\text{Reactant}} - (\sum \Delta H_{\text{comb.}})_{\text{Product}}$ $H^+ + OH^- \rightarrow H_2O$ $\Delta_N H = -57.2 \text{ kJ/mol}$ $\Delta H = \sum (B.E.)_R - \sum (B.E.)_P$ $CuSO_4(s) \rightarrow CuSO_4(aq.); \Delta H^0_{(sol.)}$ $\Delta_R H = \Delta_R H_1 + \Delta_R H_2 + \Delta_R H_3...$	$(\Delta H_f)_{\text{Product}}$ = Change in enthalpy of formation of product $(\Delta H_f)_{\text{Reactant}}$ = Change in enthalpy of formation of reactant $(\Delta H_{\text{comb.}})_{\text{Product}}$ = Change in enthalpy of combustion of product $(\Delta H_{\text{comb.}})_{\text{Reactant}}$ = Change in enthalpy of combustion of reactant $\Delta_N H$ = Heat of neutralisation $(B.E.)_R$ = Bond enthalpy of reactants $(B.E.)_P$ = Bond enthalpy of products $\Delta H^0_{(sol.)}$ = Enthalpy of solution $\Delta_R H$ = Heat of reaction

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11.	Entropy	$\Delta S = \frac{q_{\text{rev}}}{T}$ $\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$ $\Delta S_{\text{sys}} = \frac{q_{\text{sys,rev}}}{T}$ $\Delta_{\text{fusion}} S = \frac{\Delta_{\text{fusion}} H}{T_m}, \Delta_{\text{vap.}} S = \frac{\Delta_{\text{vap}} H}{T_b},$ $\Delta_{\text{sub}} S = \Delta S_{\text{fusion}} + \Delta S_{\text{vap.}}$ $\Delta_r S^\circ = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}}$	T_m = Melting point T_b = Boiling point ΔS = Change in entropy $S^\circ_{\text{Products}}$ = Standard entropy of products $S^\circ_{\text{Reactants}}$ = Standard entropy of reactants $\Delta_{\text{fusion}} H$ = Change in enthalpy of fusion $\Delta_{\text{vap}} H$ = Change in enthalpy of vaporisation $\Delta_{\text{sub}} H$ = Change in enthalpy of sublimation $\Delta_{\text{fusion}} S$ = Change in entropy of fusion $\Delta_{\text{vap}} S$ = Change in entropy of vaporisation $\Delta_{\text{sub}} S$ = Change in entropy of sublimation
12.	Gibbs free energy	$G = H - TS$	G = Gibbs energy H = Enthalpy T = Temperature S = Entropy
13.	Standard free energy of a reaction	$\Delta_r G^\circ = \sum \Delta_f G^\circ_{\text{products}} - \sum \Delta_f G^\circ_{\text{reactants}}$	$G^\circ_{\text{Products}}$ = Standard free energy of a reaction of products $G^\circ_{\text{Reactants}}$ = Standard free energy of a reaction of reactants
14.	Gibbs-Helmholtz equation	$\Delta G = \Delta H - T\Delta S$ $\Delta G^\circ = -nFE^\circ$ $\Delta G^\circ = -2.303 RT \log K$	ΔG = Change in Gibbs energy ΔH = Change in enthalpy T = Temperature ΔS = Change in entropy E° = Standard emf of cell ΔG° = Standard change in Gibbs free energy K = Equilibrium constant
15.	Gibbs energy & spontaneity:		
	(a) Spontaneous process	$\Delta G < 0$	ΔG = Change in Gibbs energy
	(b) Non-spontaneous process	$\Delta G > 0$	
	(c) Equilibrium process	$\Delta G = 0$	